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Testing Quantised Inertia on the Bending of Light by the Sun.

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Abstract

This paper applies a new cosmological theory of inertia, called quantised inertia (QI) to the dynamics of light in gravitational fields. It is shown for the first time that QI correctly predicts the observed bending of light by the Sun and the Shapiro time delay. It is also argued that QI provides a simpler and more testable mechanism than standard general relativity since Unruh radiation can be, and has been, directly observed, but a metric of space cannot.

Keywords: Cosmology: theory, Sun: general, Gravitational lensing: micro

1 Introduction

General relativity has been one of the most intensively tested theories in history. The Einstein Equivalence principle is well supported by Eotvos and clock experiments. General relativity is supported by Solar system tests such as light bending by the Sun, the Shapiro time delay, the precession of the perihelion of Mercury, the Nordvedt effect in lunar motion and frame dragging (Will, 2014). General relativity is well supported by data in all areas where the acceleration is high. However, it has failed to predict systems where the acceleration is low, for example in galaxy clusters and galaxy rotations and has required the addition of arbitrary dark matter (Zwicky, 1933, Rubin and Ford, 1970).

To address the problem in galaxy rotation McCulloch (2007, 2013) proposed a new model for inertia. When an object accelerates, say, to the right, it sees so-called Unruh radiation (Fulling, 1973, Davies, 1975, Unruh, 1976) and an information horizon forms to its left. If it is assumed that the Unruh waves have to fit into the distance between the object and the horizon (with nodes at the horizon and object) then the Unruh radiation will be damped by the horizon in the direction opposite to the acceleration and, so the Unruh radiation

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will become anisotropic, pushing the object back against its initial acceleration. This predicts inertia (McCulloch, 2013, Gine and McCulloch, 2016). It could be called a Rindler pullback. Furthermore, this model predicts that some of the Unruh radiation will also be suppressed, this time isotropically, by the distant Hubble horizon which will make this mechanism less efficient, reducing inertial mass in a new way for very low accelerations for which Unruh waves are very long (McCulloch, 2007). This model, called quantised inertia, modifies the standard inertial mass (m) as follows:

$$m_i = m\left(1 - \frac{2c^2}{|a|\Theta}\right) \tag{1}$$

where c is the speed of light, Θ is the co-moving diameter of the observable universe (which is $8.8\times10^{26}m$, Bars and Terning, 2009) and |a| is the magnitude of the acceleration of the object relative to surrounding matter. Eq. 1 predicts that for terrestrial accelerations (eg: $9.8m/s^2$) the second term in the bracket is tiny and standard inertia is recovered, but in environments where the mutual acceleration is of order $10^{-10}m/s^2$, for example at the edges of galaxies, the second term becomes larger and the inertial mass *decreases* in a new way. This modification does not affect equivalence principle tests using torsion balances since the predicted motion is independent of the mass (see Eq. 7 below).

General relativity predicts dynamics in the Solar system very well, but not in galaxies where it needs the addition of arbitrary dark matter. Quantised inertia predicts well in galaxies and in this paper it will be shown that QI can predict the light-based Solar system tests as well.

Assuming a simple Variable Speed of Light formulation (Unzicker, 2009, 2015a, 2015b) the existing empirical tests show that any successful theory must predict the following relation for the speed of light a distance r from the Sun, c_r

$$\frac{c_r}{c_\infty} = 1 - \frac{2GM}{rc^2} \tag{2}$$

This relation is derived from quantised inertia below.

2 Method

It was shown by McCulloch (2014, 2015) that Newton's gravity law can be derived simply from the Heisenberg uncertainty principle, if one allows the uncertainty in position Δx to be the distance between the two masses, r. In the derivation the masses act as information horizons. This produced classical gravity from quantum mechanics. The result of this derivation was, of course

$$F = \frac{GMm}{r^2} \tag{3}$$

where F is the gravitational force, G is the gravitational constant, M and m are the masses and r is the distance between them. To apply this dynamically we need the acceleration, so we introduce the inertial mass m_i

$$a = \frac{GMm}{m_i r^2} \tag{4}$$

We assume here that inertial mass is not the same as the gravitational mass, but is given by quantised inertia (McCulloch, 2007) as $m_i = m(1 - 2c^2/a\Theta)$ where m is the unmodified inertial or gravitational mass, c is the speed of light and Θ is the cosmic diameter. This approach correctly and simply predicts galaxy rotation without dark matter. As a result, from eq. 4, we get

$$a = \frac{GM}{(1 - {}^{2c^2}/_{a\Theta})r^2} \tag{5}$$

Multiplying through by the bracket

$$a\left(1 - \frac{2c^2}{a\Theta}\right) = \frac{GM}{r^2} \tag{6}$$

So

$$a = \frac{GM}{r^2} + \frac{2c^2}{\Theta} \tag{7}$$

The first question is what direction does the second term on the right act in? This second term represents the reduction of the Rindler pullback of QI, so it must be in a direction opposite to it, and therefore in the same direction as the initial acceleration.

Now we consider a photon of light passing by the Sun (see Figure 1). To get the velocity of the light at any one radius, we integrate its acceleration from a distance r to infinity (or at least a large distance away).

$$\int_r^\infty a dr = \int_r^\infty \frac{GM}{r^2} + \frac{2c^2}{\Theta} dr \tag{8}$$

In QI, the cosmic horizon depends upon the speed of light so the second term on the right hand side can be simplified. $2c^2/\Theta = GM/r^2$ so $\Theta = 2r^2c^2/GM$. Note that for the cosmos as a whole $2r = \Theta$ and so $c^2 = 2GM/\Theta$. This latter relation has been verified by observed cosmological paramters and has been known about for some time (Unzicker, 2015). Therefore the integral on the right hand side of Eq. 8 simplifies to

$$\int_r^\infty a dr = \int_r^\infty \left(\frac{GM}{r^2} + \frac{GM}{r^2}\right) dr \tag{9}$$

Which is

$$c[v]_r^\infty = \left[-\frac{2GM}{r}\right]_r^\infty = -\frac{2GM}{r} \tag{10}$$

So

$$c(c_r - c_\infty) = -\frac{2GM}{r} \tag{11}$$

Dividing through by c_∞^2

$$\frac{c_r}{c_\infty} = 1 - \frac{2GM}{rc_\infty^2} \tag{12}$$

This formula from quantised inertia predicts the Solar light deflection and Shapiro timing results that have been used as direct tests of general relativity and provides an alternative, and much simpler, mechanism.

3 Discussion

The following is an attempt to explain in an intuitive way. Standard Newtonian physics, which is derivable from quantum mechanics (see McCulloch, 2014, 2015) causes photons to bend around the Sun because the Sun behaves like a horizon damping the Unruh field in its vicinity so that the photon moves towards the void in that Unruh field. Here, quantised inertia doubles the normal effect of gravity because the cosmic horizon seen by the photon shrinks massively and that halves its inertial mass.

Note that the derivation here is only valid for photons, as for them $\Theta = 2c^2/a$. For matter or stars orbiting around the centre of galaxies the expression is $\Theta = 2v^2/a = 2c^2r/v_{orbit}^2 \sim 10^{26}m$ as assumed in previous papers on quantised inertia. It will need a detailed study to see if this still fits the data exactly.

This paper is part of a broader campaign to show that dynamics can be explained by a simpler and more testable scheme than general relativity. Quantised inertia is more deeply testable since Unruh radiation (vacuum fluctuations) can be directly observed. As proof of this, see Lynch et al. (2021) who have directly observed the expected acceleration-induced Unruh spectrum directly by looking at positrons decelerating in crystals. In contrast, the metric of general relativity cannot be directly seen only indirectly inferred.

Another way to detect Unruh radiation would be to use what we might call a Casimir probe. The normal Casimir effect should be enhanced if the Casimir cavity is in an accelerating environment. The acceleration would have to be considerable, however.

4 Conclusions

In this paper the theory of quantised inertia is applied to the dynamics of light in gravitational fields.

For photons the diameter of the cosmic horizon is not constant but is $\Theta = 2c^2/a$.

Accounting for this, quantised inertia correctly predicts the bending of light by the Sun and the Shapiro time delay.

It is argued that QI provides a simpler and more testable mechanism than general relativity.

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Data Availability Statement

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Statements & Declarations

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Figures

